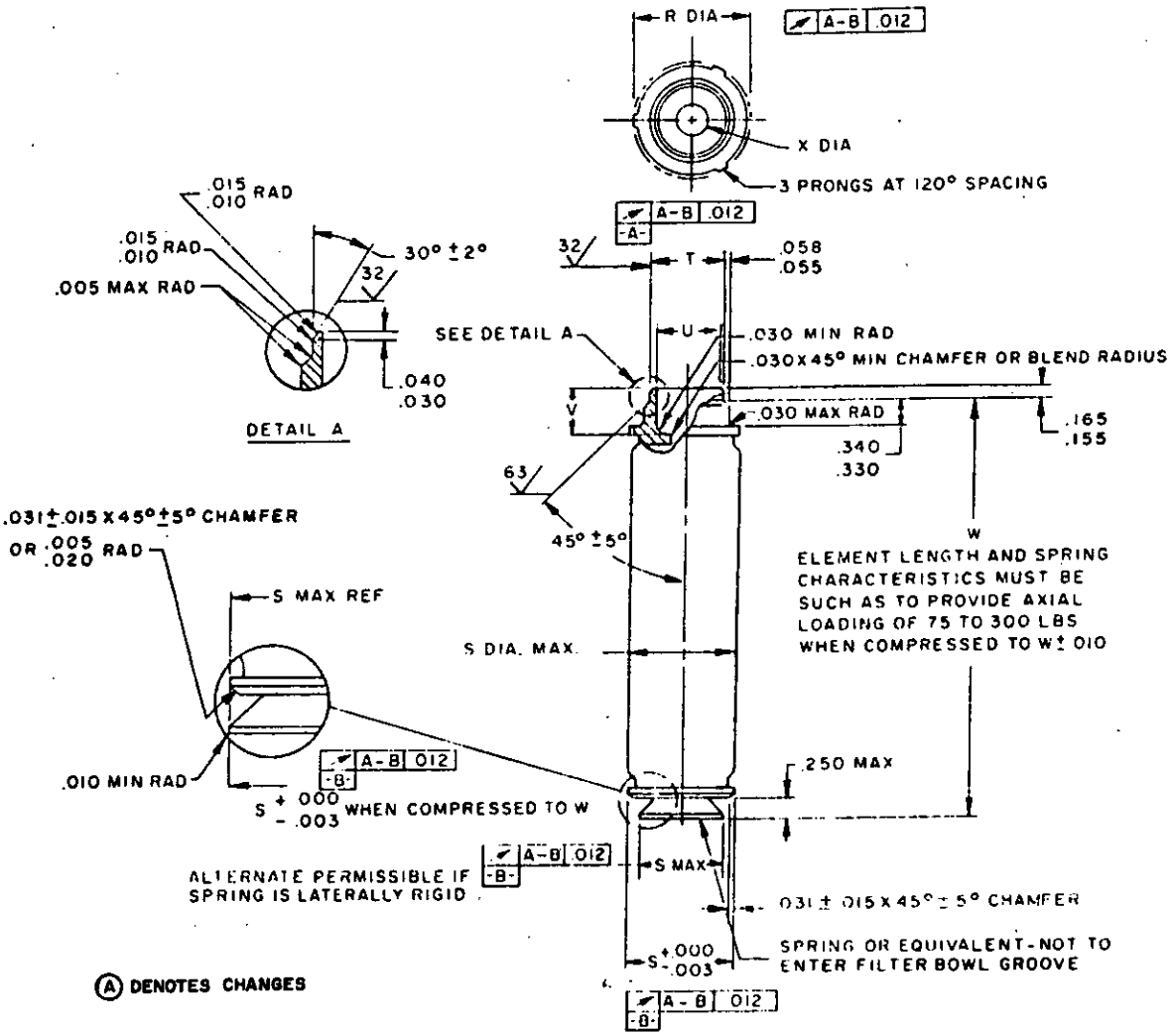
MILITARY SPECIFICATION SHEET

FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3000 PSI, 5 MICRON ABSOLUTE, AV8 AND A7 AIRCRAFT, 3.5 GPM

This specification is approved for use by the Naval Air Systems Command Department of the Navy.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.



(A) DENOTES CHANGES

FSC 1650

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NOTES: 1. Dimensions in inches unless otherwise specified. See above Figure and Table A.

2. Tolerances, decimals ± .010, angles ±1°.

3. Finish all over 100/ except as noted. Surface texture in accordance with ASA B46.1 - 1962 (ANSI).

4. Material shall be in accordance with the procurement specification.

TABLE A

DASH NO.	RATED MAX FLOW GPM	R +.000 -.010	S	T ±.001	U MIN	V MIN	W	X MIN	MAX WT LBS
-1	3.5	1.250	1.137	.748	.670	.875	3.625	.480	0.31

REQUIREMENTS:

Pressure Drop: Maximum pressure drop at rated flow across the filter element shall not exceed 12 psid.

Degree of Filtration: The filter element shall remove 100 percent of all contaminant larger than 5 microns and 95 percent of all test contaminant. The filter efficiency shall be a minimum of 95 percent.

Element Collapse: The filter element shall withstand a differential pressure of 4500 psid.

Cold Start: The element shall withstand 3000 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop shall not exceed 12 psid following the cold start test.

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Maximum Particle Passed: The maximum particle passed shall be no greater than 5 microns.

Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 CPM maximum. (A)

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
500 cycles at 3000 psid pressure drop

Part Number: The part number consists of the basic number of this specification sheet, and a dash number taken from Table A.

Example:

M8815/18-1

TABLE I. CONTAMINANTS

Dash No.	Flow Rate (gpm)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup		Migration Analysis Value	
								Maximum Allowable (mgm)	
		F-9 Beads (gms)	AC Dust (gms)	F-9 Beads (gms)	Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)	After Cold Start - 4.7.2.3.1	After Vibration 4.7.2.8.3.2
-1	3.5	0.090	0.090	0.025 (A)	0.025	0.15	1.40	3.3	0.75

Intended Use: The filter element covered by this specification sheet is intended for use in the model AV8 and A7 aircraft operating with hydraulic fluid conforming to MIL-H-5606 at nominal operating pressures of 3000 psi.

Preparing activity:
Navy - AS

(Project No. 1650-N052-10)

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DOCUMENT IDENTIFIER AND TITLE MIL-F-8815/18A(AS) FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3000 PSI, 5 MICRON ABSOLUTE, AV8 AND A7 AIRCRAFT, 3.5 GPM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO (If "Yes", in what way?)

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